

Total No. of Questions : 8]

SEAT No. :

PB2543

[Total No. of Pages : 2

[6263] 442

B.E. (Computer Engineering) (Honours in Data Science)
ARTIFICIAL INTELLIGENCE FOR BIG DATA ANALYTICS
(2019 Pattern) (Semester -VIII) (410503)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6 and Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

Q1) a) What are the components of a feed forward neural network? Explain with the help of neat sketch. **[6]**

b) Difference between linear and nonlinear neural networks? **[6]**

c) What is Recurrent neural network? Explain in detail. **[8]**

OR

Q2) a) Explain Gradient descent algorithm that is used to train the neural networks. **[6]**

b) Explain fundamental of neural networks in big data. **[6]**

c) How the backpropagation algorithm works? Explain with suitable example. **[8]**

Q3) a) Explain YARN : Yet Another Resource Negotiator. **[4]**

b) Explain HDFS (Hadoop Distributed File System) with the help of neat sketch. **[6]**

c) What is Apache Spark? Explain how it overcomes the problem of Speed of computation. **[8]**

OR

Q4) a) Explain Apache Hadoop ecosystem with suitable diagram. **[4]**

b) Explain Spark : In-Memory data processing. **[6]**

c) Explain Map/Reduce functionality of Hadoop. How it is use for data aggregation. **[8]**

P.T.O.

- Q5) a)** Explain the following : **[6]**
- i) Data Integration
 - ii) Data Transformation methods
- b) What are the different key characteristics of a Data Warehouse? **[6]**
- c) What is Hive Big Data and its Benefits? **[4]**

OR

- Q6) a)** Explain in detail Data mining functionalities? **[6]**
- b) Explain Scalable Machine Learning on Big Data using Spark. **[6]**
- c) Write a short note on : Data warehousing. **[4]**
- Q7) a)** List and explain the applications of NLP. **[6]**
- b) What is object detection? Explain any one application where in object detection is performed. **[6]**
- c) Explain Feature Extraction Techniques in NLP. **[4]**

OR

- Q8) a)** What is sentiment analysis Explain with application in detail. **[6]**
- b) Explain the applications of computer vision. **[6]**
- c) What is the objective of Natural Language Processing? **[4]**

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